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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech. (CBCS) Degree Examinations

December 2025

EC3106-1 – CONNECTED DEVICES SECURITY

(For Minor Degree)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which is an example of IoT in healthcare?
A) Smart Grids B) FitBit wearables
C) Vehicle-to-vehicle communication D) Smart meters
2. Which phase of IoT lifecycle involves cloud APIs and backend services?
A) Device Implementation B) Service Implementation
C) Deployment D) Retirement
3. In deployment phase of IoT lifecycle, which aspect becomes most critical?
A) Device authorization and monitoring B) Compiler selection
C) Circuit Design D) Hardware Sourcing
4. Which IoT protocol works on publish/subscribe model?
A) ICMP B) FTP
C) HTTP D) MQTT
5. Which messaging protocol is UDP-based and designed for constrained devices?
A) MQTT B) FTP
C) CoAP D) XMPP
6. Bluetooth Low Energy achieves power saving by:
A) Using IPv6 B) Using optical signal
C) Dynamic key exchange D) Defaulting to sleep mode
7. Which of the following is an IoT threat example?
A) DoS on IoT gateway B) SQL injection in databases
C) Device Spoofing D) All the options are correct
8. Which is an example of a physical IoT attack?
A) JTAG port tampering B) SQL injection
C) Cross-site scripting D) Packet sniffing
9. Why is security training important for IoT administrators?
A) To increase memory capacity B) To configure devices securely and handle incidents
C) To write software faster D) To replace hardware
10. Which of the following is a secure configuration practice for IoT devices?
A) Locking unused ports B) Allowing unsigned firmware
C) Using default passwords D) Sharing keys
11. Which key was the default ZigBee key exploited by attackers?
A) ZigBeeDefault01 B) ZigBeeAlliance09
C) IoTAlliance07 D) DefaultKey123
12. What does ABAC in IoT access management stand for?
A) Attribute-Based Access Control B) Advanced Blockchain Access Control
C) Automated Biometric Authentication Control D) Anonymous Based Access Control

13. Which anomaly might indicate an IoT security breach?
 A) Regular sensor updates B) Encrypted communication
 C) Sudden spike in activity D) Valid firmware
14. White-box penetration testing assumes:
 A) Full internal knowledge B) No prior knowledge of system
 C) Only physical access D) Use of random guessing
15. What is the role of cryptography in IoT systems?
 A) To speed up sensors B) To increase device range
 C) To manage device batteries D) To ensure confidentiality, integrity, authentication, and non-repudiation
16. Which algorithm is commonly used for symmetric encryption in IoT?
 A) DES B) RSA
 C) ECC D) SHA-256
17. Virtual endpoint threats in IoT cloud include:
 A) Unused port locking B) Insecure brokers and SQL injection
 C) Physical theft of devices D) Radio jamming
18. Which AWS IoT feature maintains device state even when offline?
 A) CloudTrail B) Thing Shadow
 C) Kinesis D) CloudWatch
19. Which Azure IoT Hub feature ensures per-device security?
 A) Per-device authentication and tokens B) Default key usage
 C) Single password for all devices D) Lack of event monitoring
20. Cisco Fog Computing primarily improves IoT by:
 A) Increasing latency B) Processing data at the edge
 C) Eliminating gateways D) Using only central servers

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the importance of cross-industry collaboration in IoT security. Give examples of how threats in one sector can affect others.	8	L*2	1	1
	b) Discuss with the example of attacker who wishes to accomplish the overarching goal of re-directing an Unmanned Aircraft Systems (UAS), that is, a drone, while in flight.	8	L2	2	1
2.	a) With suitable examples, describe current IoT applications in at least three areas: energy industry, transportation, and healthcare.	8	L2	1	1
	b) Compare fault trees and attack trees. How can merging both analyses improve the security of CPS?	8	L2	2	1
3.	a) What is the IoT device lifecycle? Describe its three phases (device implementation, service implementation, and deployment).	8	L2	1	1
	b) Explain any four common IoT attack types with examples.	8	L2	2	1
Unit – II					
4.	a) Discuss Public Key Infrastructure (PKI) and its application in IoT.	8	L2	3	1
	b) Explain the need for Verification and Validation (V&V) in IoT systems. How do they ensure system security?	8	L2	4	1
5.	a) Explain secure boot in IoT devices.	8	L2	3	1
	b) Discuss the significance of security training for both IoT system users and administrators. Give examples of training topics.	8	L2	4	1
6.	a) Define lightweight cryptography. Why is it necessary in IoT?	8	L2	3	1

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| b) | What are the common secure configuration practices for IoT devices and gateways? Explain with examples. | 8 | L2 | 4 | 1 |
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Unit – III

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| 7. | a) | Explain the following terms with respect to Network Security Terminology: Cryptography, Hacking, Access Control and Authorization. | 8 | L2 | 5 | 1 |
| | b) | Describe the main threats to IoT systems from a cloud perspective. | 8 | L2 | 5 | 1 |
| 8. | a) | Explain the operational security with block diagram. | 8 | L2 | 5 | 1 |
| | b) | Discuss AWS IoT and its features for secure IoT cloud deployment. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
